

ADITYA SAXENA

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Education

Indian Institute of Technology Delhi

Delhi, India

BTech in Engineering & Computational Mechanics, Minor in Computer Science

May 2026

GPA: 8.85/10.0 (Dept Rank 5)

Awards: Summer Undergraduate Research Award (2024), Semester Merit Award (Sem-II 23-24)

Selected Coursework: NLP, Advanced Reinforcement Learning, Probabilistic ML, Database Management Systems, Information retrieval and Web Search, ML, Digital Image Processing, Analysis and Design of Algorithms, Discrete Mathematics, Probability and Stochastic Processes, Linear Algebra & Differential Equations

Technical Skills

- **Programming Languages** - Python, C++, Javascript
- **ML/DL Libraries:** PyTorch, OpenCV, NumPy, Pandas, Matplotlib
- **Frameworks/Tools:** Huggingface, Gymnasium, MongoDB, Express.js, React.js, Node.js, Git

Professional Experience

IBM Research Labs — (AI Research Intern)

May 2025 – July 2025

- Implemented a state-driven multi-agent LLM workflow using Finite State Machines (FSMs) for robust orchestration of Earth Observation tasks, improving execution reliability over single-agent baselines.
- Designed a modular LoRA-based SFT pipeline for the orchestrator and specialist agents to learn state transitions and structured tool invocation, improving correctness rate and success rate by 12-15%.
- **Best Code Repository** Runners Up award during internship at IBM Research Labs.

Projects

Multilingual CoT Distillation

April 2026

- Distilled CoT reasoning from Qwen2.5-7B into Qwen2.5-1.5B and Llama-3.2-1B students across 5 languages.
- Countered Indic token fertility with English-pivot self-consistency (k=4) + target-language rendering; raised usable Tamil/Kannada trace yield by 20% via STaR-style gold-conditioned rationalization.
- Eliminated an $9\times$ gradient skew towards Indic scripts with per-example length normalization and added script-adaptation pretraining to reduce byte-fallback corruption.

Cross-Lingual Relation Extraction with Decoder-Only LMs

March 2026

- Built a LoRA-tuned Qwen2.5-1.5B relation classifier for 5 languages. Used a staged curriculum of language-adaptive pretraining on unlabelled Indic corpora followed by task tuning on upsampled Indic data.
- Reframed generative extraction as label-ID prediction under greedy decoding to eliminate hallucinated labels, trained with interleaved LM and task loss to gain coverage of all four Indian languages.
- Designed a training-free in-context learning pipeline on Llama-3.1-8B using a two-block prompt with a byte-identical ontology prefix for vLLM prefix caching.

Adv. Reinforcement Learning

March-April 2026

- Implemented NPG, TRPO, and CPI from scratch in PyTorch, building TRPO's conjugate-gradient solver, Fisher-vector products, backtracking line search for a Gaussian policy on Hopper-v4 & HalfCheetah-v4.
- Implemented PPO-Lagrangian from scratch for constrained MDPs, using dual reward and cost critics with separate GAE estimators, supporting three actor objectives alongside two Lagrange multiplier update rules.
- Implemented Implicit Q-Learning from scratch in PyTorch using expectile regression for the value function and advantage-weighted policy extraction.

Decoder-only Transformer

Dec 2025

- Implemented a decoder-only Transformer from scratch in PyTorch, including multi-head causal self-attention, sinusoidal positional encodings, residual connections, and LayerNorm.
- Trained on TinyStories Dataset using FastTest embeddings, mixed-precision (AMP), gradient accumulation, cosine LR schedule and perplexity-based evaluation.
- Built an efficient autoregressive generation pipeline with KV-cache, temperature & top-k sampling, and beam search, enabling fast text decoding and coherent story generation.